SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology / College of Natural and Applied Sciences

Department of Informatics and Information Technology (DIIT)

BIT 111: MICROCOMPUTERS AND APPLICATIONS

TAKEHOME HOMEWORK ASSIGNMENT 4 (TOPIC 4)

Theme: FUNCTIONS & TYPES OF OPERATING SYSTEMS, DOS COMMANDS & PROCESS MANAGEMENT

Course Instructor: Madam Bertha Msuliche Lebalwa (MSc)
Curriculum Coverage: Subtopic 4.1 (OS Evolution & Categories), Subtopic 4.2 (DOS Internal & External Commands), Subtopic 4.3 (Windows/Linux System Management)
Assessment Weight: Coursework Component (Official Scheme: Assignments 5%, Tests 15%, Practical Report 10%, Practical Test 30%, UE 40% - subject to instructor review)
Submission Deadline: End of Week 8 by 17:00 EAT | Total Allocation: 100 Marks

INSTRUCTIONS TO CANDIDATES:
1. This assignment must be solved individually. Plagiarism, peer copying, or unreferenced AI generation will result in an automatic score of zero and disciplinary reporting under SUA Examination Regulations.
2. Answer ALL questions in Sections A, B, and C. Allocate your effort according to the mark allocations shown in brackets.
3. Show all intermediate formulas, binary conversions, architectural diagrams, command syntax, and arithmetic calculations clearly.
4. Where design questions require software configurations, provide exact command line syntax (Bash/DOS) or functional formulas (Excel) as specified.
5. Submit your completed work as a clean, stapled technical report or upload the single consolidated PDF to the SUA Virtual Learning Environment (Moodle) before the deadline.

SECTION A: FOUNDATIONAL OS THEORIES [25 MARKS]

Question 1 [5 Marks]: Explain the primary objectives of an Operating System as: (a) An Extended Machine (Abstraction Layer), and (b) A Resource Arbiter. Contrast how user applications interact with hardware directly versus through system call interfaces.

Question 2 [5 Marks]: Differentiate between Monolithic Kernels, Microkernels, and Hybrid Kernels. Discuss the trade-offs regarding execution performance, fault isolation, and memory footprint. Name one real-world operating system utilizing each kernel architecture.

Question 3 [5 Marks]: Describe the 5-state process lifecycle (New, Ready, Running, Waiting/Blocked, Terminated). Draw or trace the state transitions that occur when: (a) A running process requests a disk read, and (b) The disk controller signals an interrupt that the data has been loaded into memory.

Question 4 [5 Marks]: Define Virtual Memory and Demand Paging. What is a Page Fault? Explain the architectural function of the Memory Management Unit (MMU) and the Translation Lookaside Buffer (TLB) during address translation.

Question 5 [5 Marks]: Differentiate between Real-Time Operating Systems (RTOS) and General-Purpose Operating Systems (GPOS). Distinguish between 'Hard Real-Time' and 'Soft Real-Time' constraints using an automated agricultural combine harvester as an illustrative example.

SECTION B: PROCESS SCHEDULING & COMMAND SCRIPTS [35 MARKS]

Question 6 [12 Marks]: CPU Process Scheduling Problem:
Consider four processes arriving in the ready queue with the following arrival times and CPU burst times (in milliseconds):
- Process P1: Arrival = 0 ms, Burst = 8 ms
- Process P2: Arrival = 2 ms, Burst = 4 ms
- Process P3: Arrival = 4 ms, Burst = 9 ms
- Process P4: Arrival = 5 ms, Burst = 2 ms

(a) Draw a Gantt chart illustrating process execution under First-Come, First-Served (FCFS) scheduling. Calculate the average waiting time and average turnaround time. [4 Marks]
(b) Draw a Gantt chart illustrating process execution under Preemptive Shortest Remaining Time First (SRTF). Calculate the average waiting time and average turnaround time. [5 Marks]
(c) Compare both algorithms and explain why SRTF minimizes average waiting time but introduces the risk of process starvation. [3 Marks]

Question 7 [11 Marks]: MS-DOS Command Architecture and Batch Scripting:
(a) Explain the fundamental difference between an Internal DOS Command and an External DOS Command in terms of memory residency and file execution. [3 Marks]
(b) Classify each of the following DOS commands as Internal or External: `DIR`, `FORMAT`, `CHKDSK`, `COPY`, `XCOPY`, `MD`, `ATTRIB`, `DEL`. [4 Marks]
(c) Write an automated MS-DOS batch script (`BACKUP_AGRIC.BAT`) that performs the following sequence:
1. Clears the screen and checks if the directory `D:\SUA_DATA\2026` exists.
2. If it exists, creates a timestamped backup directory `E:\ARCHIVE_2026`.
3. Uses `XCOPY` with switches to copy all files and subdirectories (including empty ones) and verify data integrity.
4. Sets the archive attribute of all source files to read-only using `ATTRIB`.
5. Outputs a success message to a log file `E:\backup.log`. [4 Marks]

Question 8 [12 Marks]: Linux Process Management and Shell Pipeline Diagnostics:
A bioinformatics pipeline at SUA running under Ubuntu is consuming excessive CPU resources.
(a) Provide the exact Linux terminal pipeline to: Find the top 5 memory-consuming processes, display their PID, User, %MEM, and Command, and output the result to `/var/log/heavy_processes.txt`. [4 Marks]
(b) Explain the difference between `kill -15 (SIGTERM)` and `kill -9 (SIGKILL)`. Why should a systems administrator always attempt `SIGTERM` before executing `SIGKILL`? [4 Marks]
(c) Explain the concept of process 'Niceness' in Linux. What command adjusts the priority of a running process with PID 4120 to run with lower priority so it does not degrade interactive terminal responsiveness? [4 Marks]

SECTION C: APPLIED FILE SYSTEMS & ENTERPRISE ADMIN [40 MARKS]

Question 9 [15 Marks]: Enterprise File Systems Comparison: FAT32 vs NTFS vs ext4:
(a) Construct a technical comparison matrix evaluating FAT32, NTFS, and Linux ext4 across: Maximum single file size, Maximum volume size, Journaling support, File-level access control permissions (ACLs), File-level encryption, and Compression capabilities. [8 Marks]
(b) Explain why a 64 GB USB flash drive formatted with FAT32 cannot store a single 5.2 GB disk image file of a SUA digital library archive, despite having 58 GB of free space. How does formatting with exFAT or NTFS resolve this limitation? [4 Marks]
(c) What is a 'Journaling File System'? Trace the three-step sequence of journal commit, file system update, and checkpointing, explaining how journaling prevents file corruption during an abrupt power outage. [3 Marks]

Question 10 [15 Marks]: System Management and Troubleshooting: Windows Administrative Utilities:
An IT specialist at SUA is resolving multiple client workstation issues.
(a) Detail the specific administrative utility (Device Manager, Task Manager, Event Viewer, Disk Management, Services MMC, or Windows Registry) required to resolve each of the following scenarios: [6 Marks]
- Scenario 1: A newly installed PCIe network adapter displays a yellow exclamation mark and fails to obtain an IP address.
- Scenario 2: A computer boots to a black screen after an unverified driver installation; the technician needs to inspect system error logs and bugcheck codes.

- Scenario 3: An unpartitioned 2 TB secondary SATA SSD needs to be initialized with GPT and formatted as an NTFS drive labeled 'Data'.
(b) Explain the architecture of the Windows Registry. Differentiate between `HKEY_LOCAL_MACHINE` (HKLM) and `HKEY_CURRENT_USER` (HKCU). Why can incorrect registry editing cause operating system boot failure? [5 Marks]
(c) Write PowerShell cmdlets to: (i) List all stopped services that have their startup type set to Automatic, and (ii) Forcefully start the `Spooler` service. [4 Marks]

Question 11 [10 Marks]: Virtualization & Containerization within Academic Computing:
(a) Differentiate between Type-1 (Bare-Metal) Hypervisors (e.g., VMware ESXi, Proxmox VE) and Type-2 (Hosted) Hypervisors (e.g., VirtualBox, VMware Workstation). Explain which is preferred for university server datacenters and why. [4 Marks]
(b) Compare a Virtual Machine (VM) with a Docker Container in terms of operating system kernel sharing, boot time, resource overhead, and isolation boundaries. [4 Marks]
(c) Explain how containerization allows agricultural researchers at SUA to package Python machine learning models with exact library dependencies for repeatable field deployment. [2 Marks]